Informatica® Cloud Application Integration
May 2025

Generate Image with Amazon Bedrock

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Publication Date: 2025-08-31

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Preface

Use *Generate Image with Amazon Bedrock* to learn how to create images that align with the text-based descriptions provided in natural language prompts. This guide assumes that you have an understanding of the Amazon Bedrock Connector and File Connector concepts.

CHAPTER 1

Introduction to Generate Image with Amazon Bedrock recipe

The Generate Image with Amazon Bedrock recipe is based on REST and SOAP APIs. The recipe creates images that align with the text-based descriptions provided in the natural language prompts.

The recipe allows you to provide a user prompt, specify the image name and format, read the prompt, generate an image using the Amazon Bedrock Large Language Model (LLM), and save it in the user-specified directory.

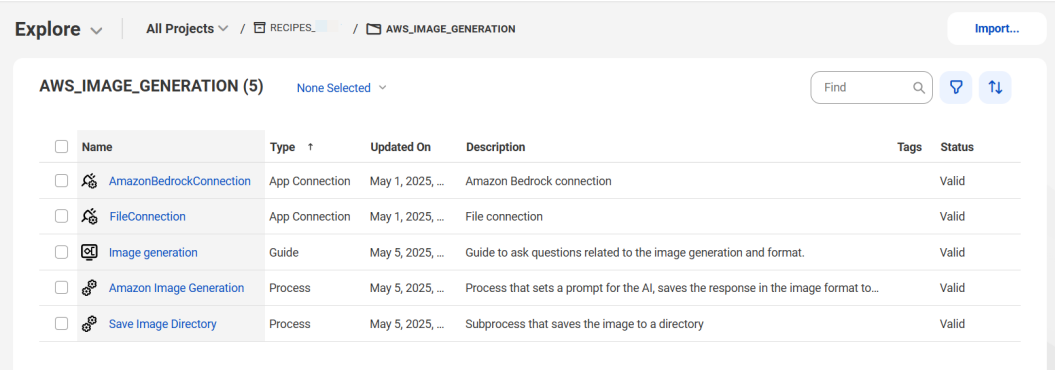
The recipe generates images in the following formats:

`.png`, `.jpeg`, `.jpg`

Generate Image with Amazon Bedrock recipe contents

The Generate Image with Amazon Bedrock recipe contains app connections, processes, and a guide.

The following image shows the assets that the Generate Image with Amazon Bedrock recipe package contains:



Generate Image with Amazon Bedrock recipe assets

The following table lists the assets that the Generate Image with Amazon Bedrock recipe package contains:

Asset Name	Asset Type	Description
AmazonBedrockConnection	App connection	Amazon Bedrock connection.
FileConnection	App connection	File connection.
Save Image Directory	Process	Subprocess that saves the image to a directory.
Amazon Image Generation	Process	Process that sets a prompt for the AI, saves the response in the image format to a user-specified directory. The user-specified directory must exist within the Secure Agent installation directory.
Image generation	Guide	Guide to ask questions related to the image generation and format.

CHAPTER 2

Using the Generate Image with Amazon Bedrock recipe

To use the Generate Image with Amazon Bedrock recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the Amazon Bedrock connection.
3. Configure and publish the File connection.
4. Configure and publish the processes.
5. Publish and run the guide.

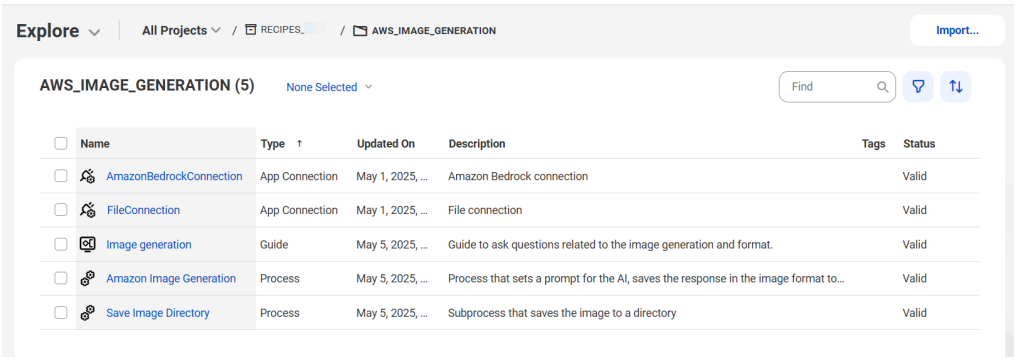
Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Generate Image with Amazon Bedrock** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Amazon Bedrock connection

Configure the access key, secret key, and region to connect to Amazon Bedrock, and then publish the AmazonBedrockConnection connection.

1. Open the **AmazonBedrockConnection** connection.
2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
AccessKey	The access key to access Amazon Bedrock.
SecretKey	The secret key to access Amazon Bedrock.
Region	The region that hosts Amazon Bedrock.

3. Save and publish the connection.

Step 3. Configure and publish the File connection

Configure the file directory and specify the action to be taken if a file already exists, and then publish the FileConnection connection.

1. Open the **FileConnection** connection.
2. From the **Runtime Environment** list, select the Secure Agent.
3. On the **Event Targets** tab, in the **Directory** field, enter the directory to store the files. In the **File Exists** field, select **TryRename**.
4. Save and publish the connection.

Step 4. Configure and publish the processes

Configure the Amazon LLM model version and publish the process.

1. To publish the **Save Image Directory** process, click **Actions** in the row that contains the process and select **Publish**.
2. Open the **Amazon Image Generation** process.
3. Optionally, on the **Temp Fields** tab of the Start step, update values in the following fields:
   - In the **Model_ID** field, update the version of the Amazon LLM model. By default, the model ID is set to amazon.titan-image-generator-v1.
   - In the **Image_Height** field and **Image_Width** field, update the height and width of the image. By default, the height and width are set to 1024.

4. Optionally, on the **Assignments** tab of the **Request for Invoke Model** step, update the expression to eliminate negative elements from appearing in the image, such as blurry, low quality, distracting backgrounds, unnatural colors, and shadow or lighting issues.
5. Save and publish the process.

Step 5. Publish and run the guide

You can ask questions in natural language to the Amazon Bedrock Large Language Model (LLM) related to image generation and provide image format.

1. Open the **Image generation** guide.
2. On the **Start** tab of the Start step, ensure that the **Run As** field is set to **Current User**.
3. Save and publish the guide.
4. On the **Actions** menu, click **Run**. Alternatively, you can copy the execution URL from the **Properties Details** dialog box to run the guide.
5. On the **Instructions** page, in the **User prompt to LLM** field, enter a prompt to generate an image.
6. In the **Image name with format** field, enter the image name along with the image format with which the generated image will get saved in the user-specified directory.

   Note: The recipe supports only `.png`, `.jpeg`, and `.jpg` image formats.
7. Click **Continue**.
   You will receive a response from the LLM.
8. To generate a new image, click **Restart**.
9. Enter a new prompt and image name with format and click **Continue**.

You can also use the embed code to embed the guide into an HTML document of a third-party application.